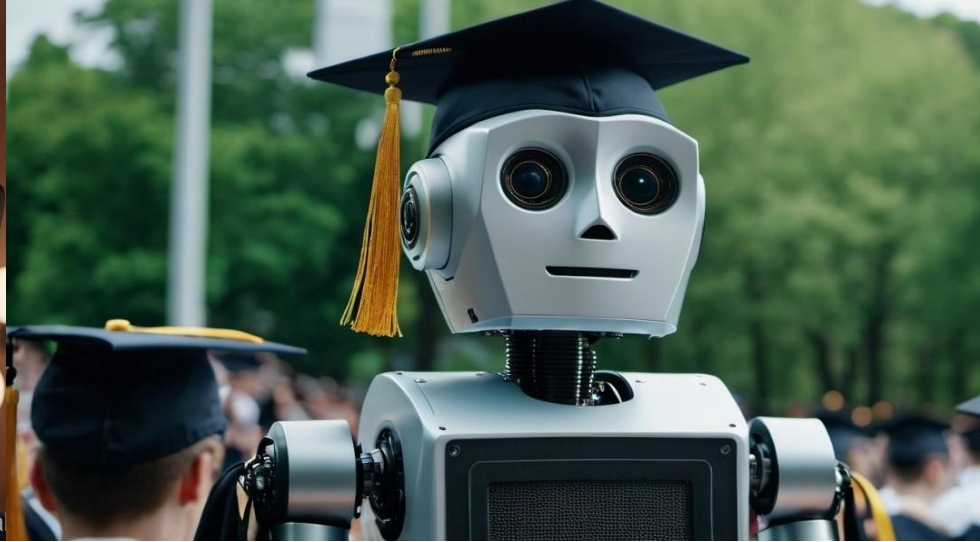




Unsupervised and Self-Supervised Learning

A brief overview of one of the greatest promises in AI

Dr. Gabriel Bertocco and Prof. Anderson Rocha

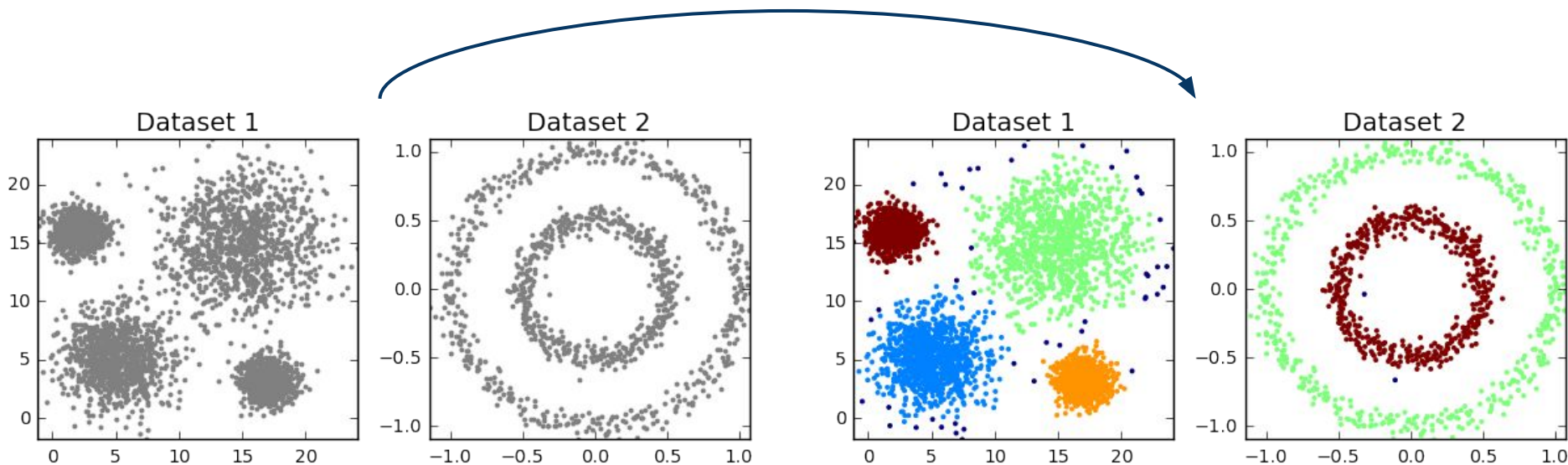
Unsupervised Learning

“...the training data consists of a set of **input vectors x without any corresponding target values**. The goal in such unsupervised learning problems may be to discover groups of similar examples within the data, where it is called **clustering**, or to determine the distribution of data within the input space, known as **density estimation**, or to project the data from a high-dimensional space down to two or three dimensions for the purpose of **visualization**”

- Christopher Bishop, PRML

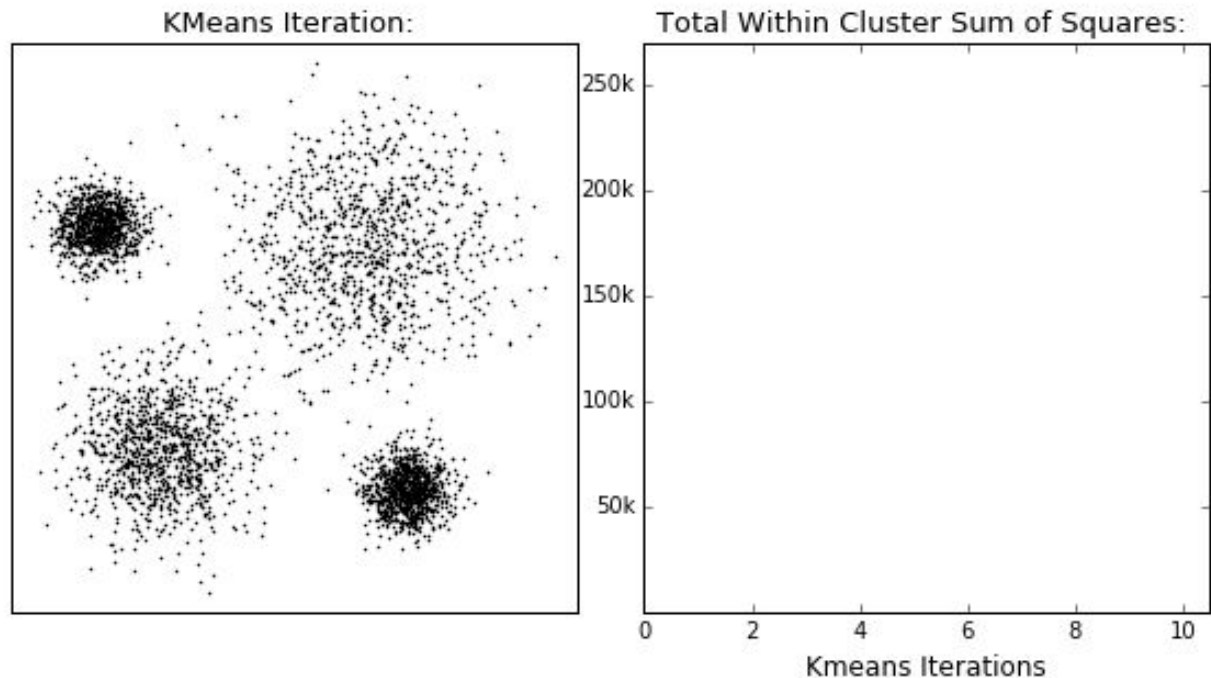
Clustering

Goal is to find meaningful groups in the data



Clustering - K-Means

Given a number K , find K clusters that minimize the Mean Squared Error to the centroids



Pros

Scalable and can be easily implemented in GPU

Incremental versions

Cons

Convex clusters

We need to define K and the centroid initialization

Doesn't detect outliers

Clustering - Agglomerative Clustering

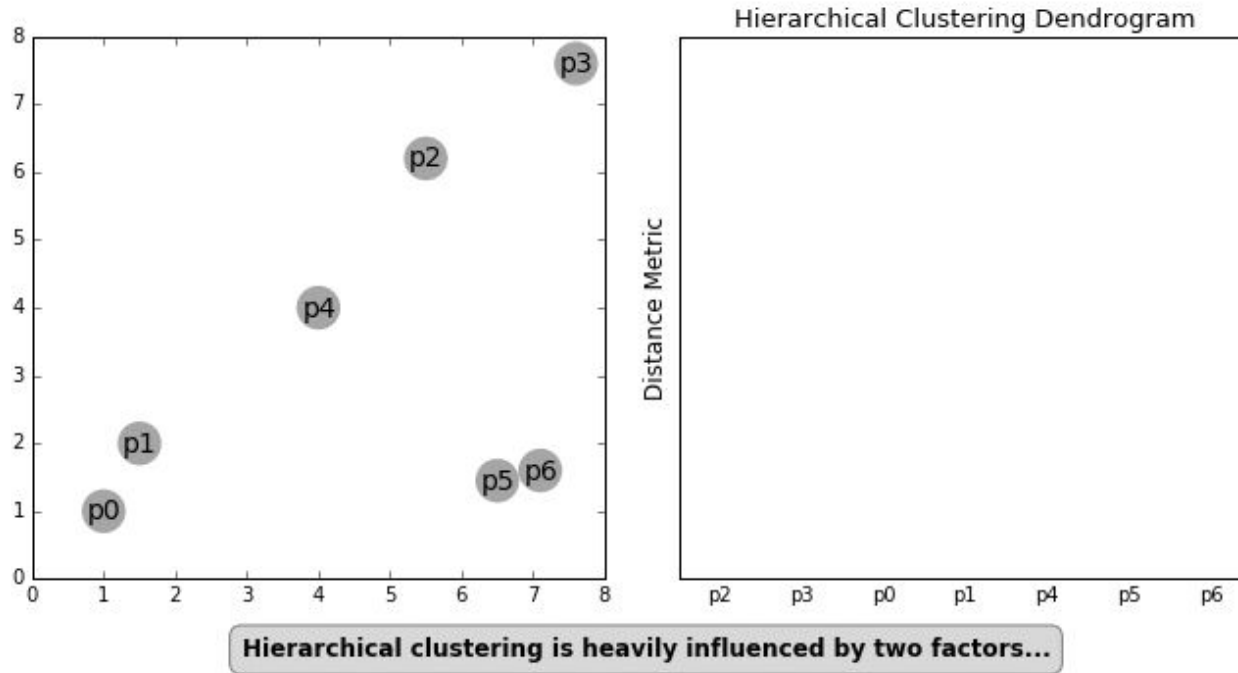
Find clusters based on the distance between points or group of points

Pros

- It works with or without choice of number of clusters
- Non-convex clusters
- Single points can be seen as outliers

Cons

- Sensible to outliers
- Unbalanced clusters
- It might be sensible to the linkage criterion



Clustering - Density-based Clustering

It considers the context (neighborhood) of each point to define clusters

Pros

It doesn't require the choice of number of clusters

Non-convex and any-shape clusters

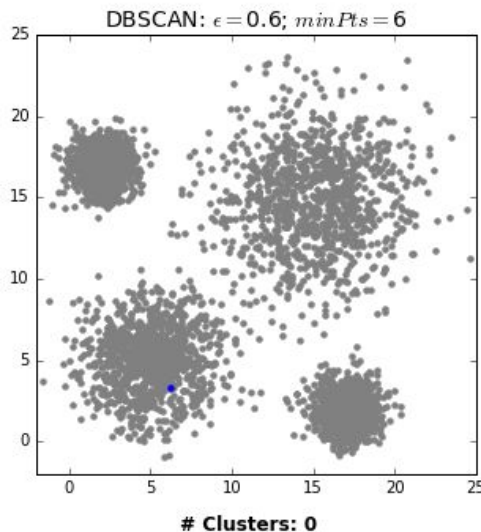
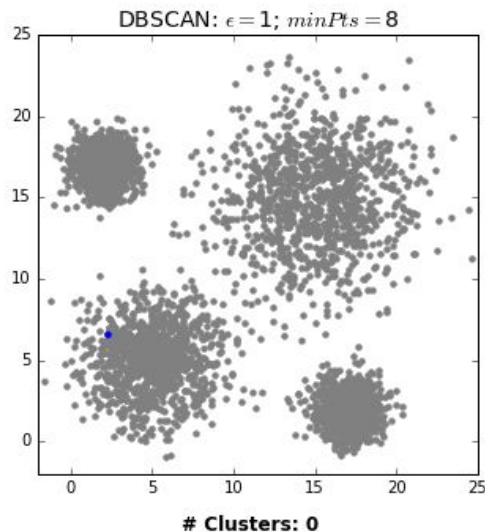
Automatically detect outliers

Some models can have fast implementations (DBSCAN)

Cons

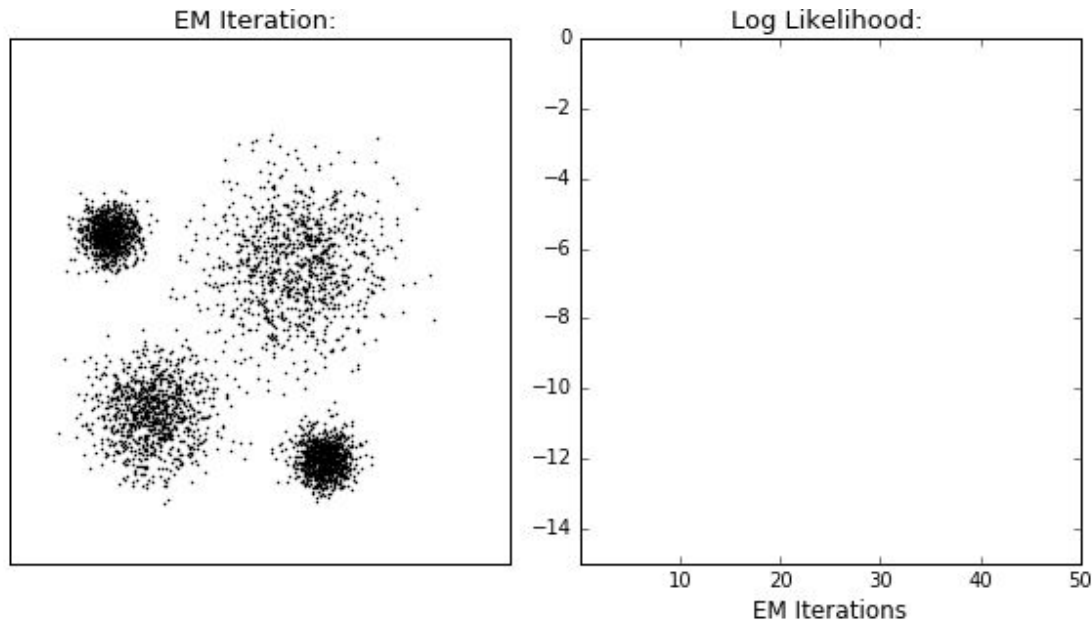
Sensible to the density parameter

It doesn't find clusters with varied densities



Clustering - Expectation Maximization

Most of clustering algorithms assign a point to single cluster (hard assignment). What if we assume the points belong to two or more clusters with varied probabilities (soft assignment)?



Pros

The points are not assigned to a specific cluster

It works well with uncertainty

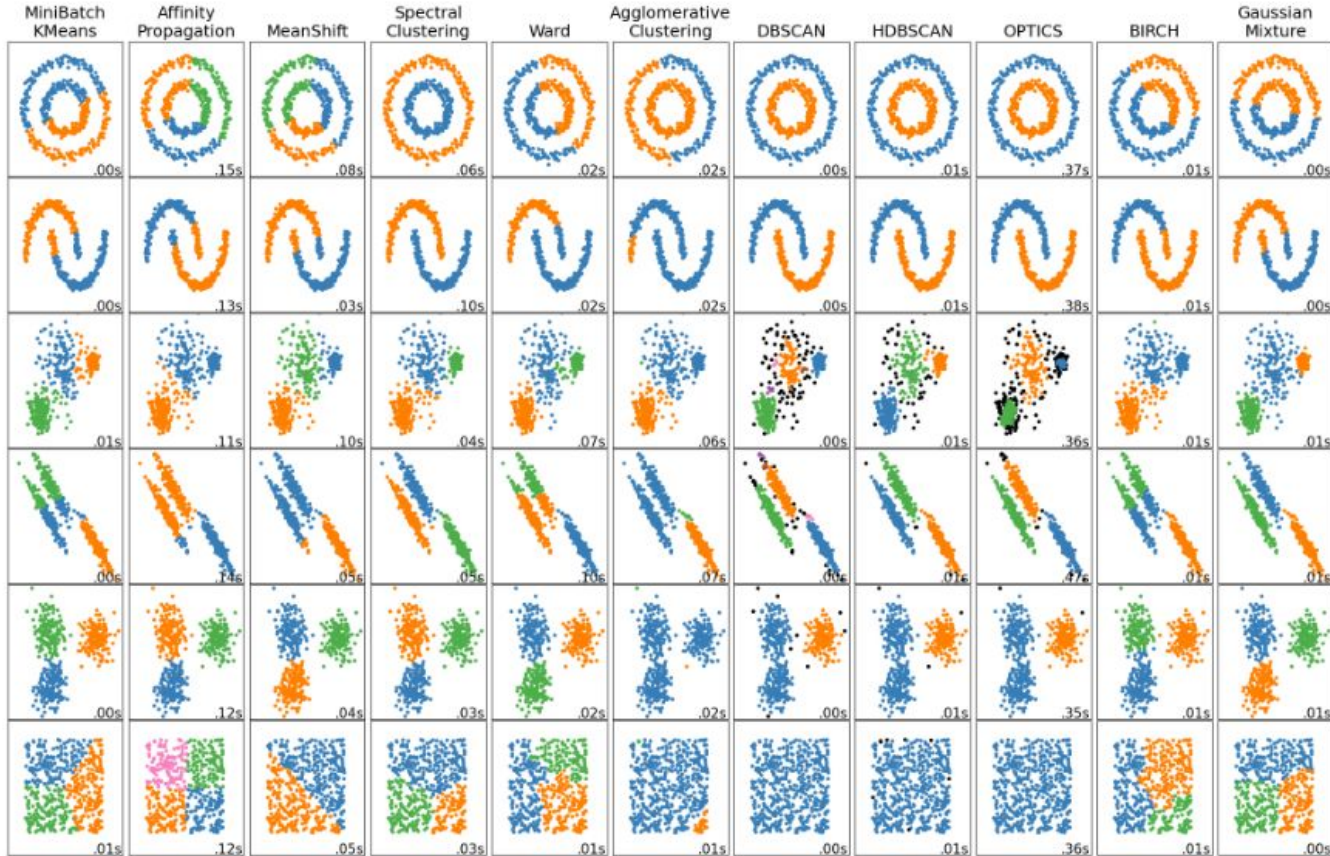
Cons

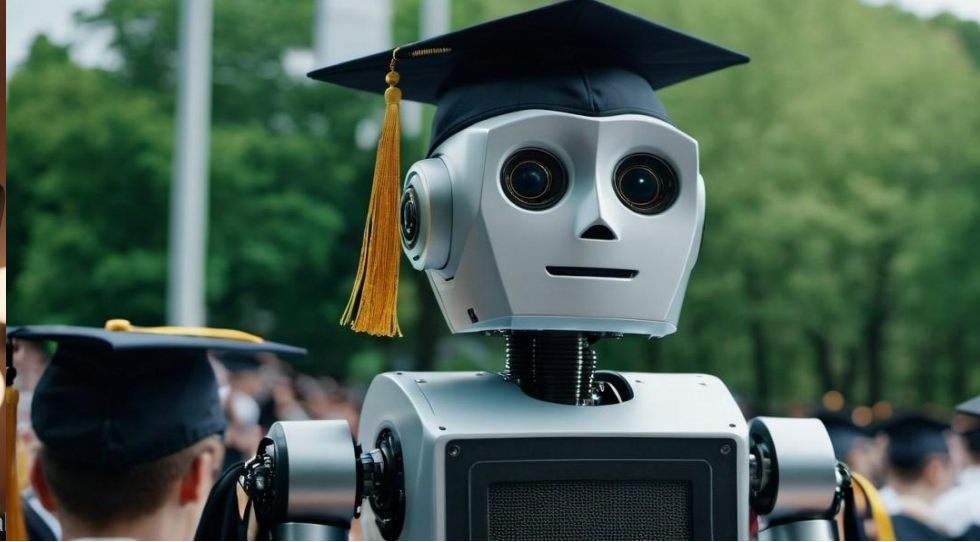
Assumes a gaussian distribution of points in the clusters

It requires the selection of the number of clusters

No guarantees of convergence 7

Clustering - Comparison of algorithms





Unsupervised and Self-Supervised Learning

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Self-Supervised Learning

“Self-supervised learning is a subset of unsupervised learning methods. Self-supervised learning refers to learning methods in which ConvNets (and general models) are explicitly trained with **supervisory signals that are generated from the data itself (self-supervision)** by leveraging its structure”

- **Longlong Jing and Yingli Tian** in "*Self-supervised visual feature learning with deep neural networks: A survey.*" published in the IEEE transactions on pattern analysis and machine intelligence

Self-Supervised Learning

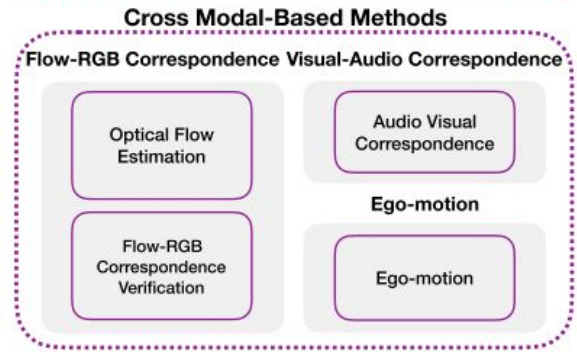
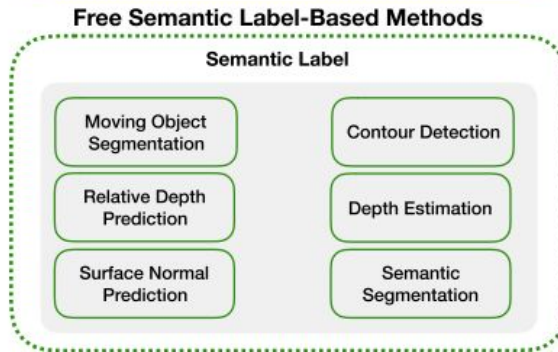
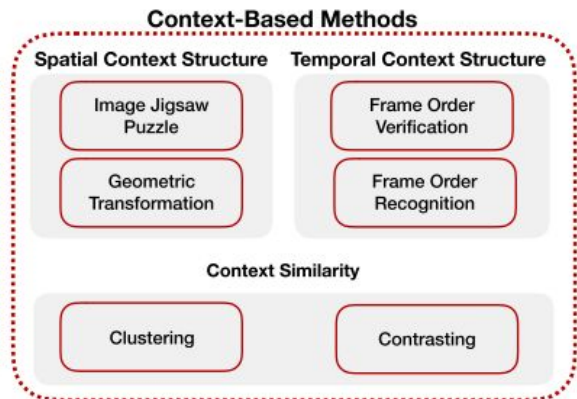
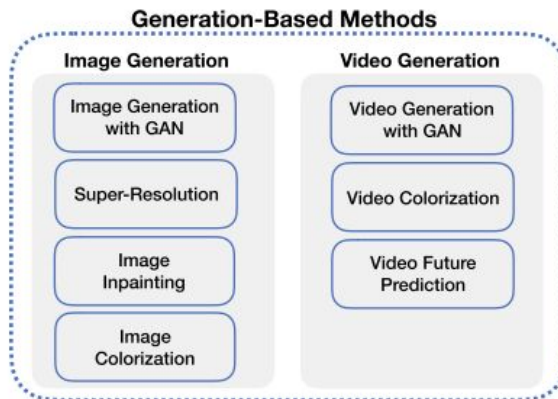
“Self-supervised learning, dubbed “the dark matter of intelligence”, is a promising path to **advance** machine learning”

- **Meta and collaborators** in *A Cookbook of Self-Supervised Learning*

“Self-supervised learning obtains **supervisory signals from the data itself**, often leveraging the underlying structure in the data”

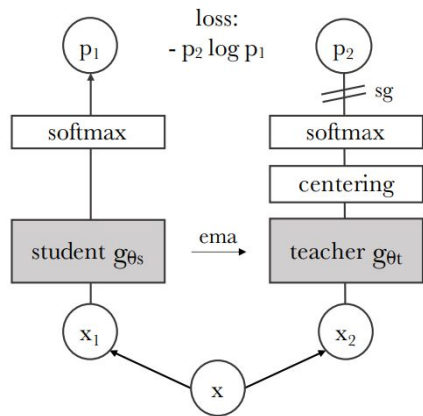
- **Meta** in the article *Self-supervised learning: The dark matter of intelligence*

Self-Supervised Learning

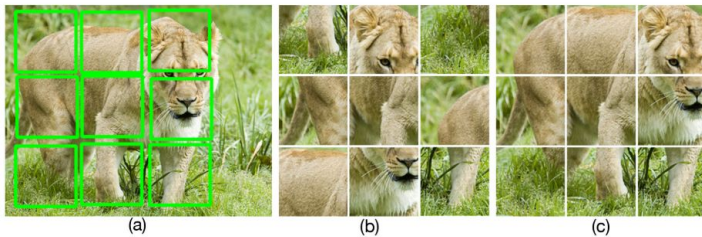


Self-Supervised Learning

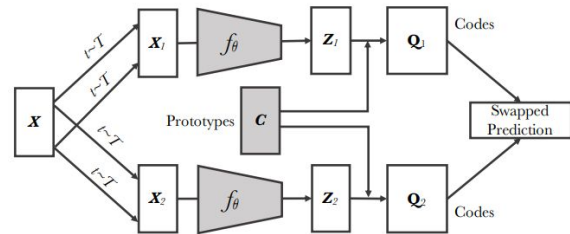
Instance-based Learning



Generation-based Learning



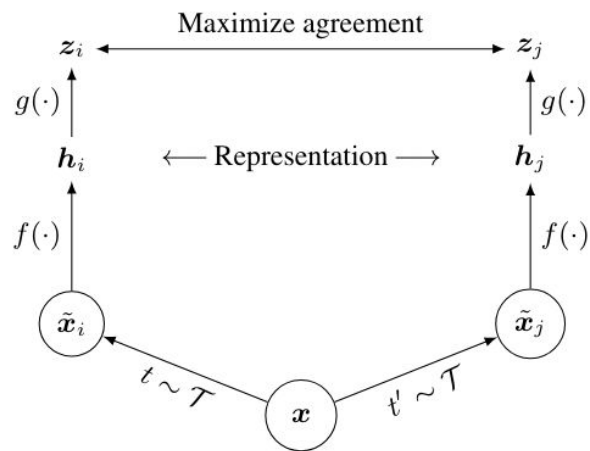
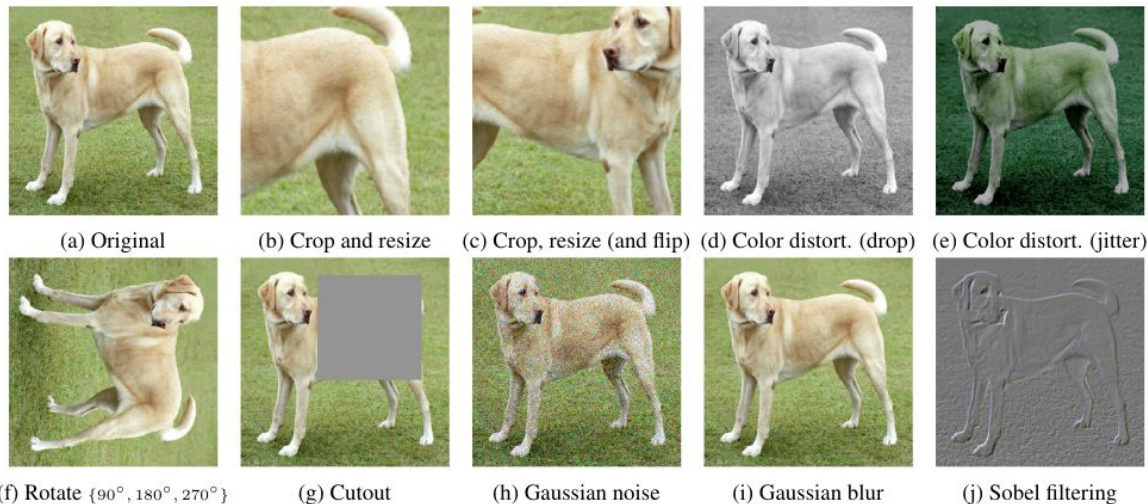
Clustering-based Learning



Not suitable for problems with **low intra-class similarity** and **high inter-class similarity**

Proper self-supervised solutions must be designed to address this complex requirement

Instance-based SSL - SimCLR



Instance-based SSL - SimCLR

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
<i>Linear evaluation:</i>												
SimCLR (ours)	76.9	95.3	80.2	48.4	65.9	60.0	61.2	84.2	78.9	89.2	93.9	95.0
Supervised	75.2	95.7	81.2	56.4	64.9	68.8	63.8	83.8	78.7	92.3	94.1	94.2
<i>Fine-tuned:</i>												
SimCLR (ours)	89.4	98.6	89.0	78.2	68.1	92.1	87.0	86.6	77.8	92.1	94.1	97.6
Supervised	88.7	98.3	88.7	77.8	67.0	91.4	88.0	86.5	78.8	93.2	94.2	98.0
Random init	88.3	96.0	81.9	77.0	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

Table 8. Comparison of transfer learning performance of our self-supervised approach with supervised baselines across 12 natural image classification datasets, for ResNet-50 (4×) models pretrained on ImageNet. Results not significantly worse than the best ($p > 0.05$, permutation test) are shown in bold. See Appendix B.8 for experimental details and results with standard ResNet-50.

Instance-based SSL - Momentum Contrast (MoCo)

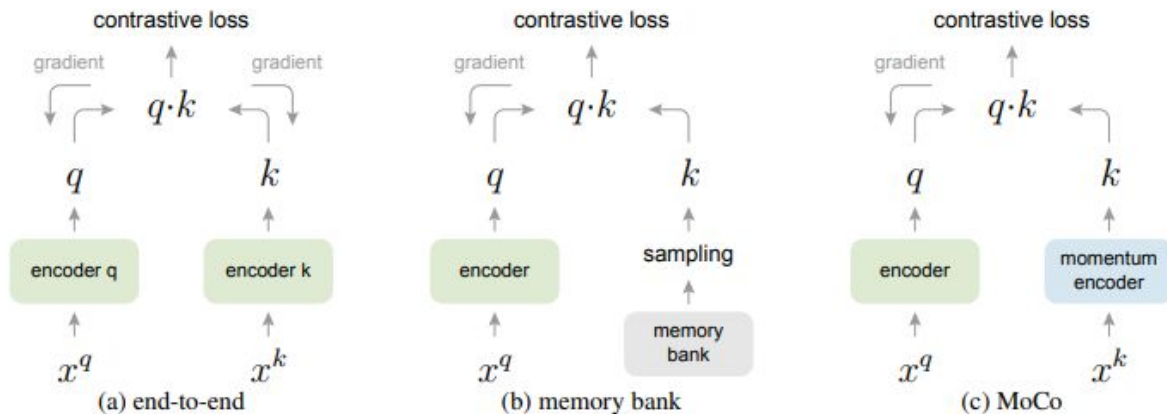


Figure 2. **Conceptual comparison of three contrastive loss mechanisms** (empirical comparisons are in Figure 3 and Table 3). Here we illustrate one pair of query and key. The three mechanisms differ in how the keys are maintained and how the key encoder is updated. (a): The encoders for computing the query and key representations are updated *end-to-end* by back-propagation (the two encoders can be different). (b): The key representations are sampled from a *memory bank* [61]. (c): **MoCo** encodes the new keys on-the-fly by a momentum-updated encoder, and maintains a queue (not illustrated in this figure) of keys.

Instance-based SSL - Momentum Contrast (MoCo)

method	architecture	#params (M)	accuracy (%)
Exemplar [17]	R50w3×	211	46.0 [38]
RelativePosition [13]	R50w2×	94	51.4 [38]
Jigsaw [45]	R50w2×	94	44.6 [38]
Rotation [19]	Rv50w4×	86	55.4 [38]
Colorization [64]	R101*	28	39.6 [14]
DeepCluster [3]	VGG [53]	15	48.4 [4]
BigBiGAN [16]	R50	24	56.6
	Rv50w4×	86	61.3
<i>methods based on contrastive learning follow:</i>			
InstDisc [61]	R50	24	54.0
LocalAgg [66]	R50	24	58.8
CPC v1 [46]	R101*	28	48.7
CPC v2 [35]	R170* _{wider}	303	65.9
CMC [56]	R50 _{L+ab}	47	64.1 [†]
	R50w2× _{L+ab}	188	68.4 [†]
AMDIM [2]	AMDIM _{small}	194	63.5 [†]
	AMDIM _{large}	626	68.1 [†]
MoCo	R50	24	60.6
	RX50	46	63.9
	R50w2×	94	65.4
	R50w4×	375	68.6

Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

```

# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
    x_q = aug(x) # a randomly augmented version
    x_k = aug(x) # another randomly augmented version

    q = f_q.forward(x_q) # queries: NxK
    k = f_k.forward(x_k) # keys: NxK
    k = k.detach() # no gradient to keys

    # positive logits: Nx1
    l_pos = bmm(q.view(N,1,C), k.view(N,C,1))

    # negative logits: NxK
    l_neg = mm(q.view(N,C), queue.view(C,K))

    # logits: Nx(1+K)
    logits = cat([l_pos, l_neg], dim=1)

    # contrastive loss, Eqn. (1)
    labels = zeros(N) # positives are the 0-th
    loss = CrossEntropyLoss(logits/t, labels)

    # SGD update: query network
    loss.backward()
    update(f_q.params)

    # momentum update: key network
    f_k.params = m*f_k.params+(1-m)*f_q.params

    # update dictionary
    enqueue(queue, k) # enqueue the current minibatch
    dequeue(queue) # dequeue the earliest minibatch

```

bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.

Instance-based SSL - Dino

$$P_s(x)^{(i)} = \frac{\exp(g_{\theta_s}(x)^{(i)} / \tau_s)}{\sum_{k=1}^K \exp(g_{\theta_s}(x)^{(k)} / \tau_s)}, \quad \min_{\theta_s} H(P_t(x), P_s(x)), \quad \min_{\theta_s} \sum_{x \in \{x_1^g, x_2^g\}} \sum_{\substack{x' \in V \\ x' \neq x}} H(P_t(x), P_s(x'))$$

<https://github.com/facebookresearch/dino>

Caron, Mathilde, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. "Emerging properties in self-supervised vision transformers." In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9650-9660. 2021.

Instance-based SSL - Dino

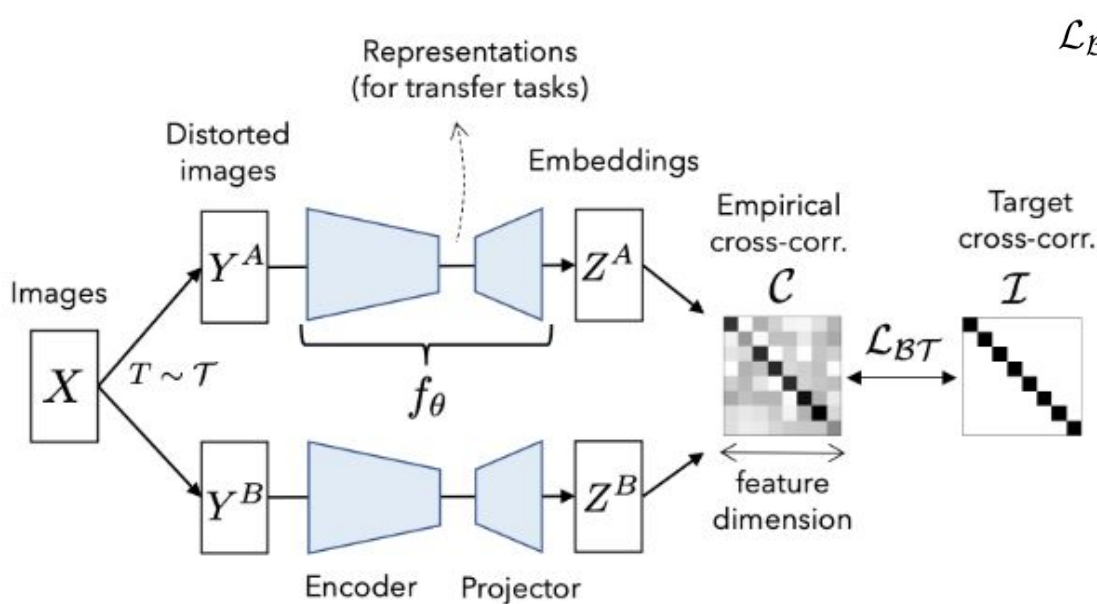


Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

<https://github.com/facebookresearch/dino>

Caron, Mathilde, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. "Emerging properties in self-supervised vision transformers." In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9650-9660. 2021.

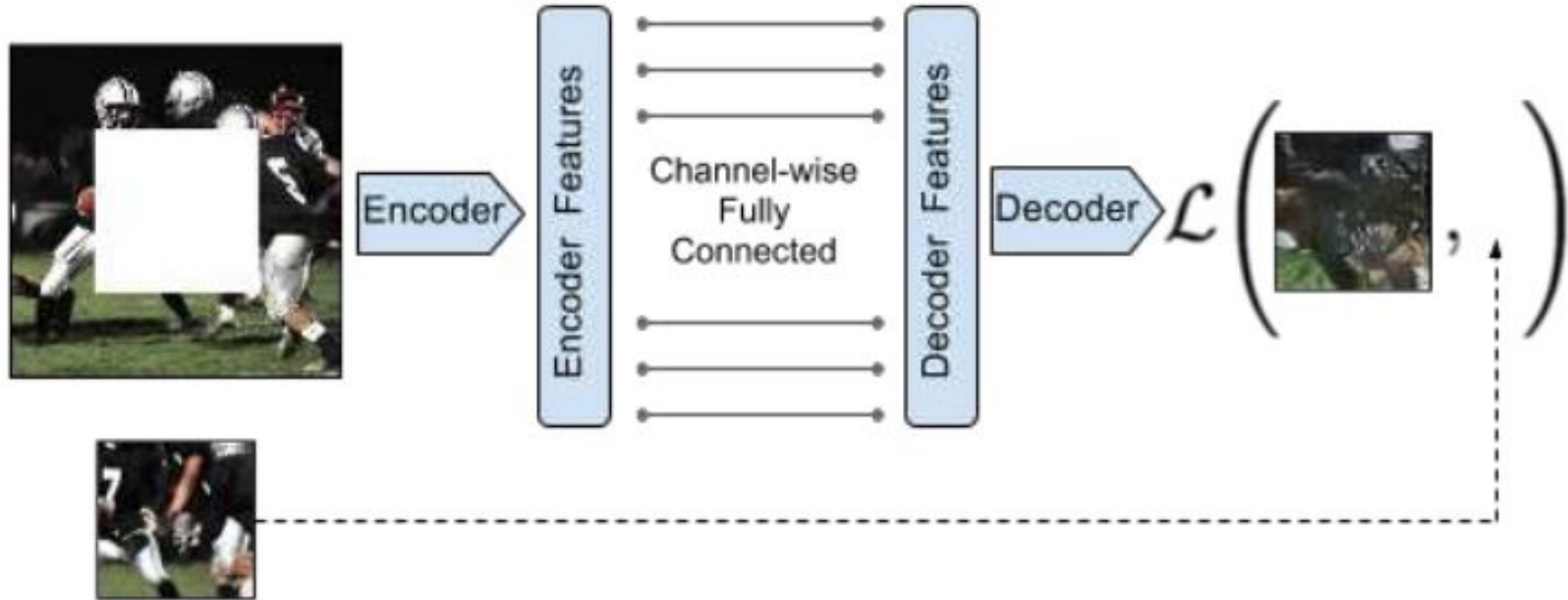
Instance-based SSL - Barlow Twins



$$\mathcal{L}_{\mathcal{B}\mathcal{T}} \triangleq \underbrace{\sum_i (1 - C_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} C_{ij}^2}_{\text{redundancy reduction term}}$$

$$C_{ij} \triangleq \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}}$$

Generation-based SSL - Inpainting



Generation-based SSL - Inpainting



(a) Input Context

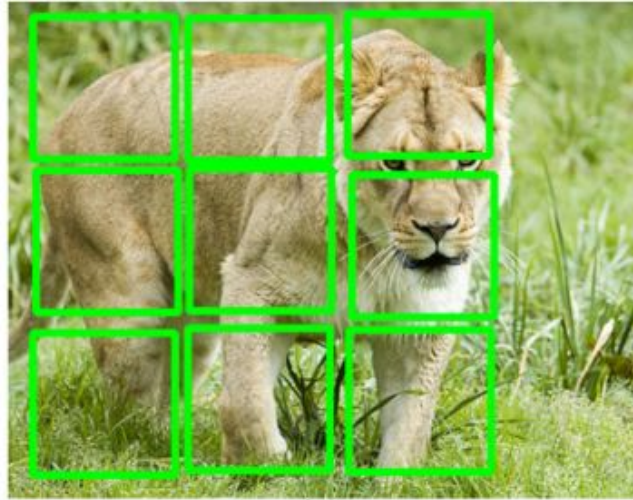


(b) Human Artist



(c) Network Prediction

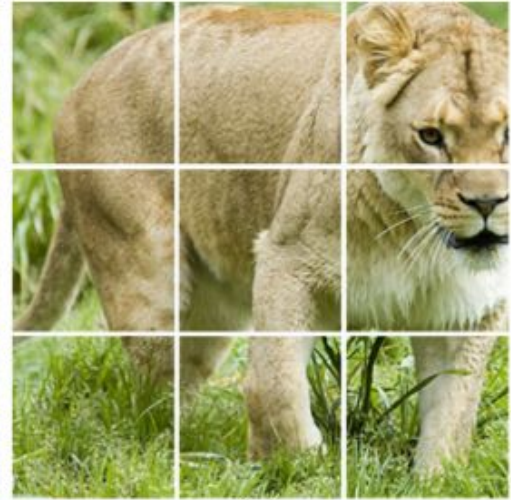
Generation-based SSL - Jigsaw Puzzle



(a)

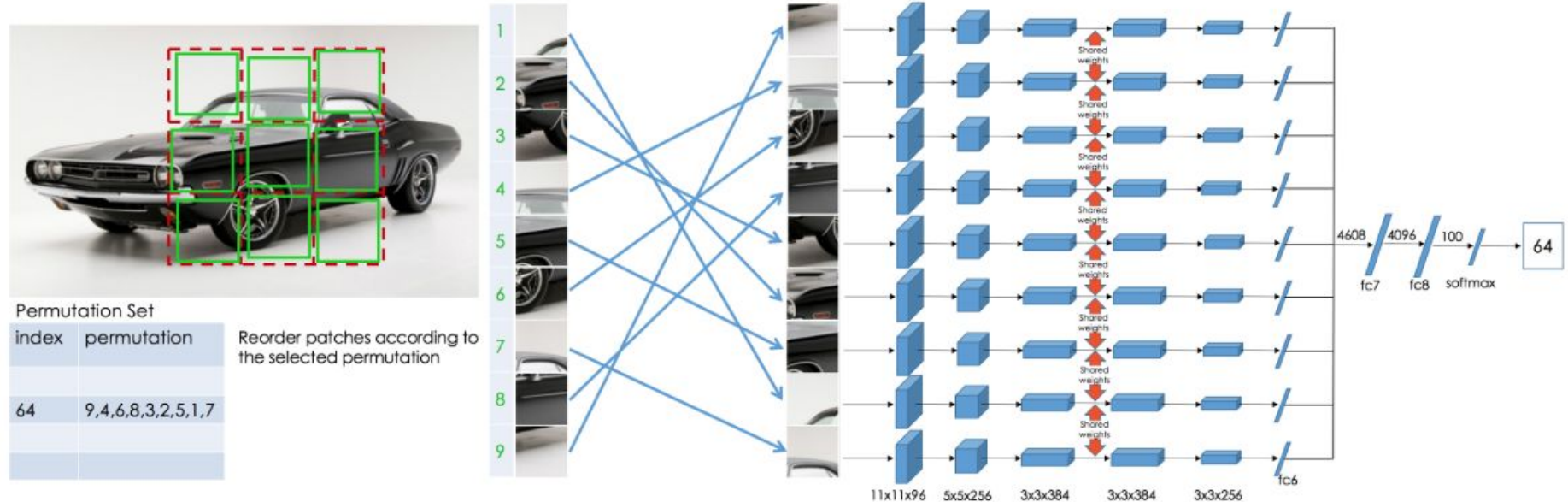


(b)

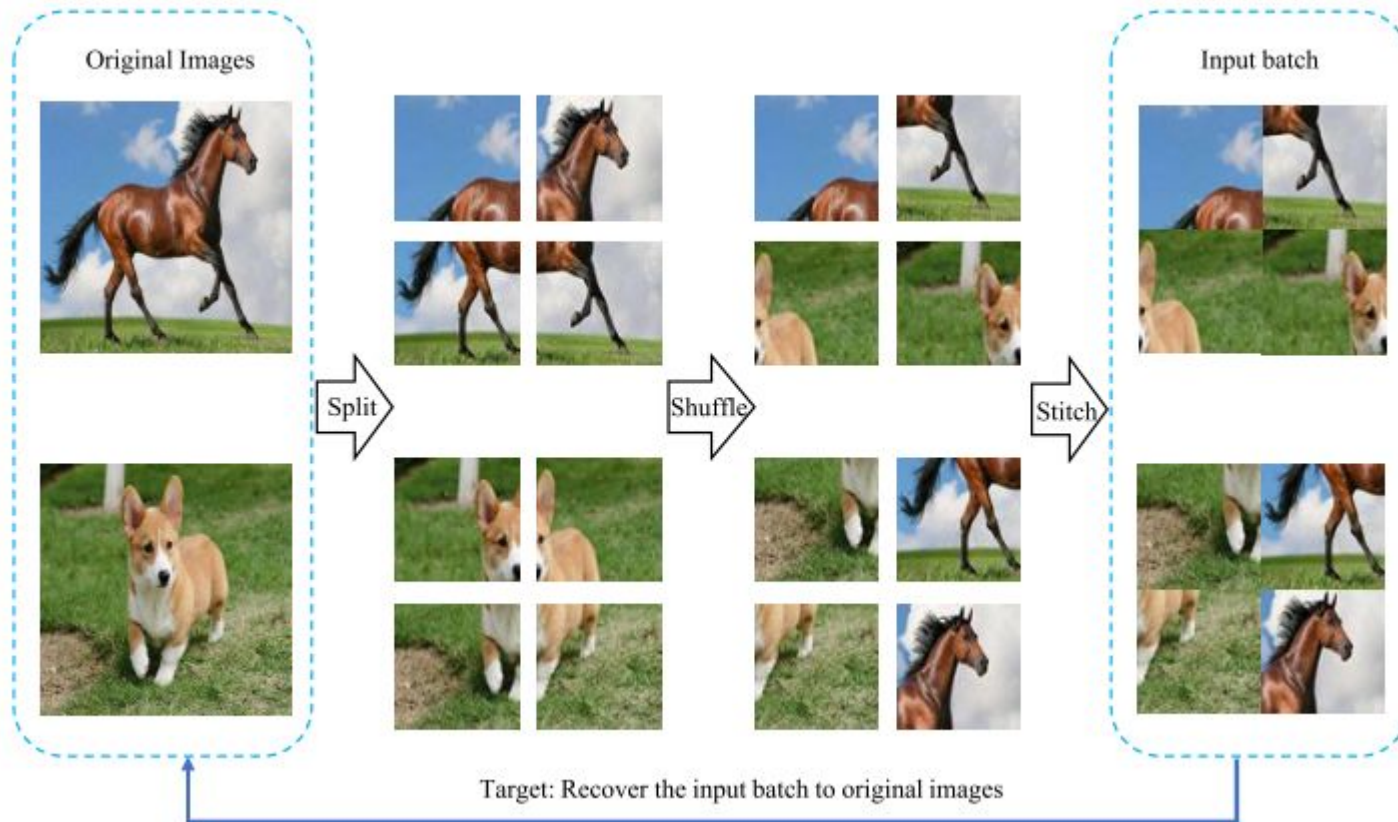


(c)

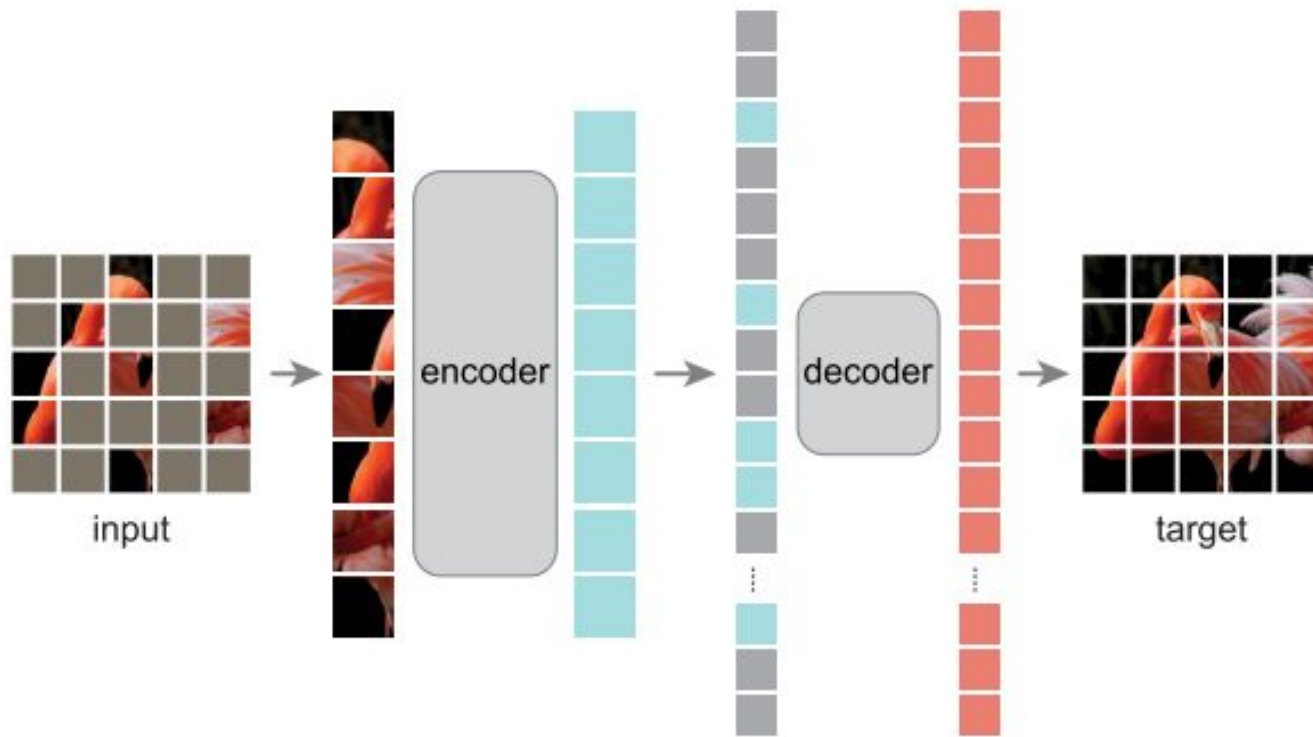
Generation-based SSL - Jigsaw Puzzle



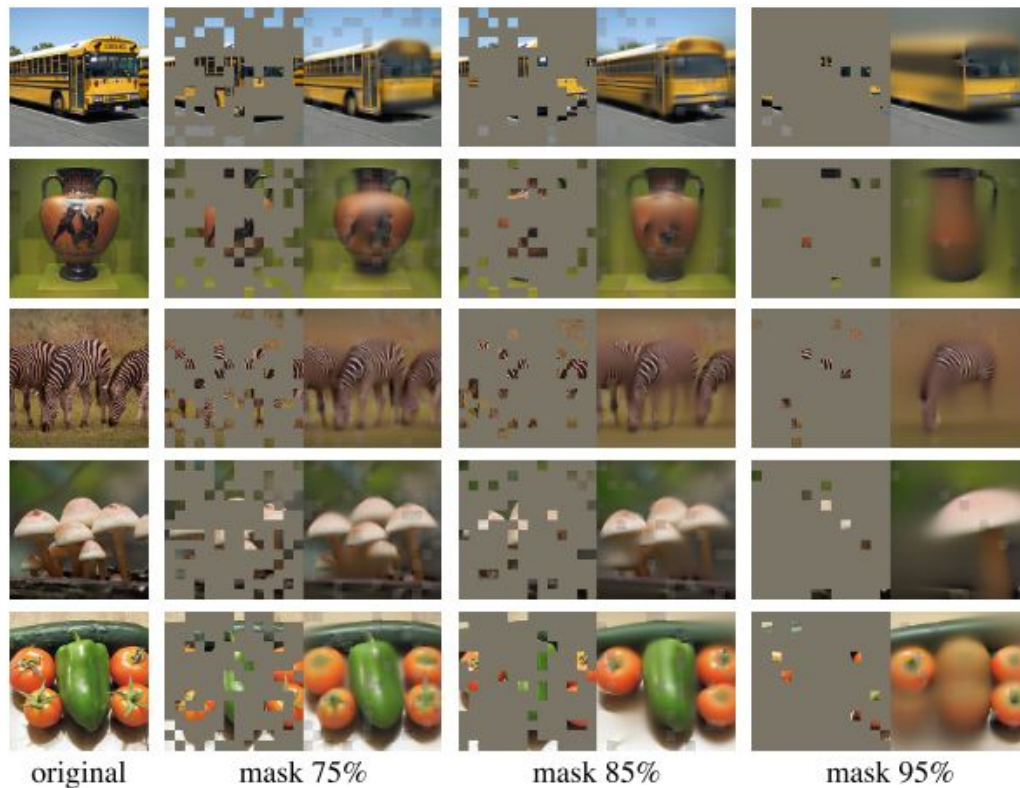
Generation-based SSL - Two-images Jigsaw Puzzle



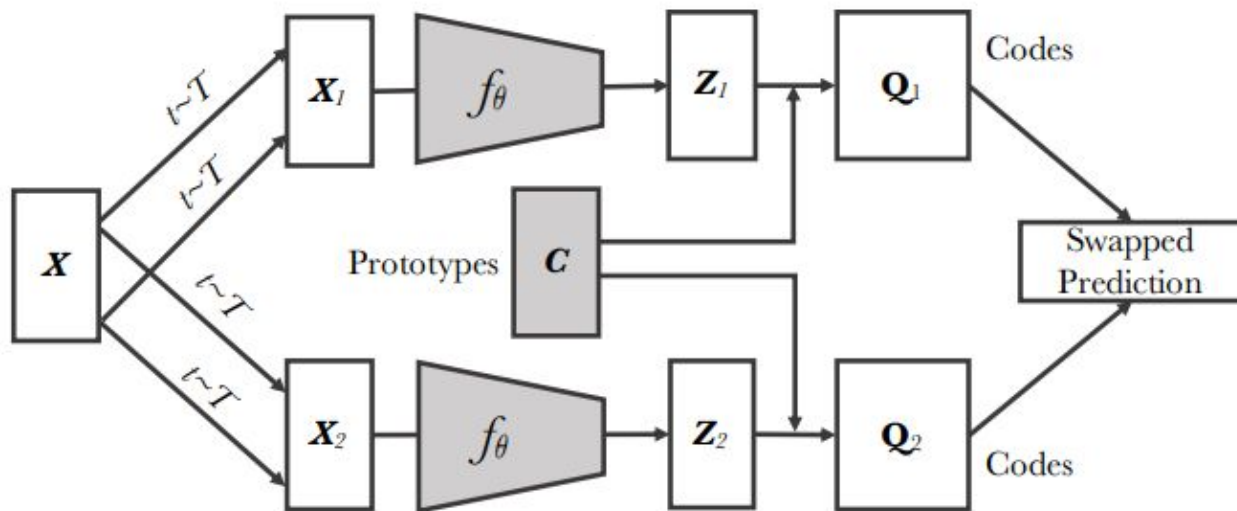
Generation-based SSL - MAE



Generation-based SSL - MAE



Clustering-based SSL - SwAV

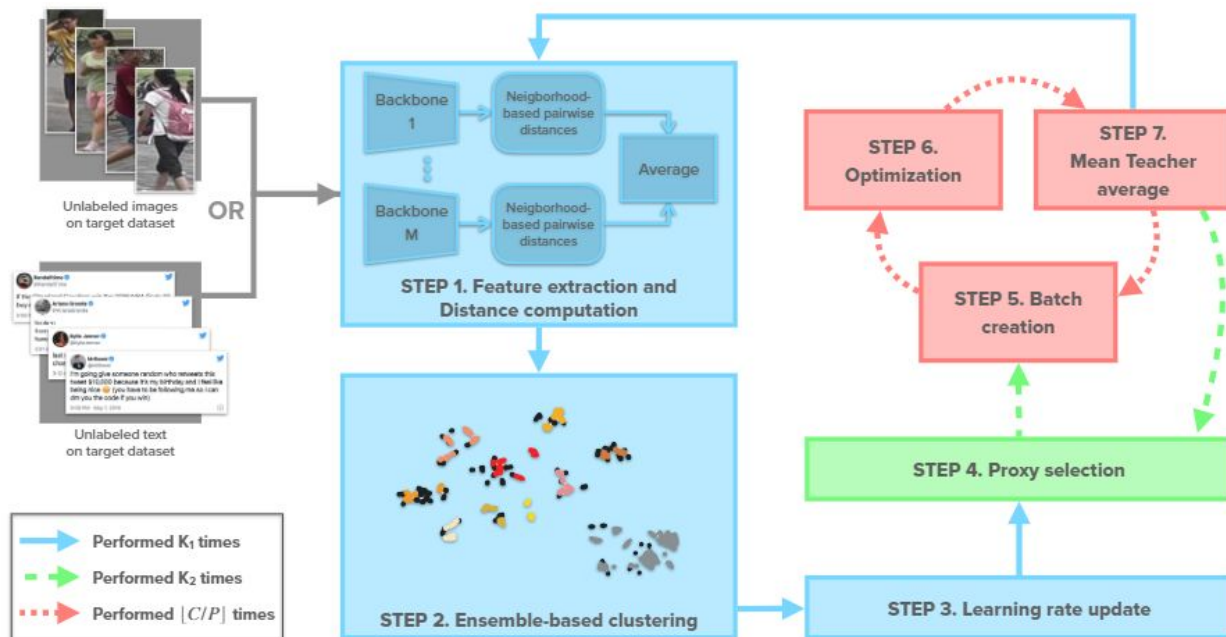


$$\ell(\mathbf{z}_t, \mathbf{q}_s) = - \sum_k \mathbf{q}_s^{(k)} \log \mathbf{p}_t^{(k)}, \quad \text{where} \quad \mathbf{p}_t^{(k)} = \frac{\exp\left(\frac{1}{\tau} \mathbf{z}_t^\top \mathbf{c}_k\right)}{\sum_{k'} \exp\left(\frac{1}{\tau} \mathbf{z}_t^\top \mathbf{c}_{k'}\right)}$$

Clustering-based SSL - Leveraging Ensembles

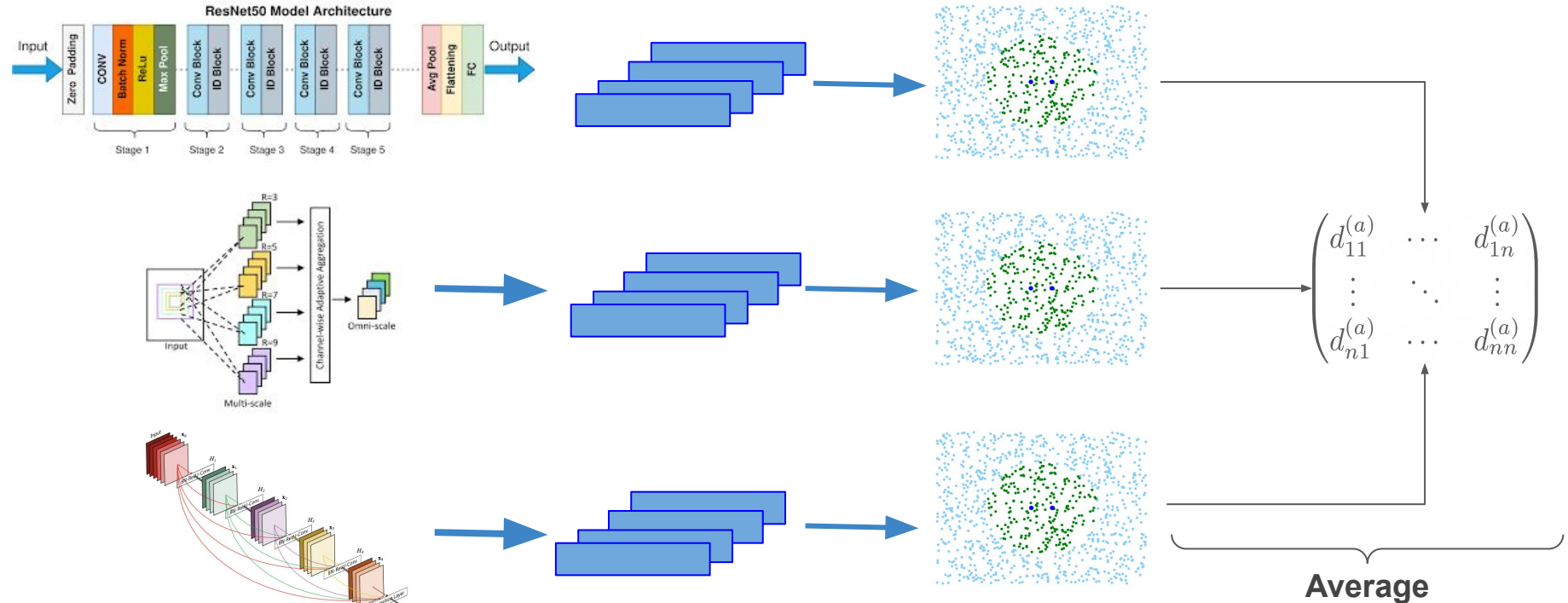
Our second pipeline **does not assume** any problem-related **pretraining or camera information**.

It is extendable to further modalities (e.g., Authorship Verification in NLP)

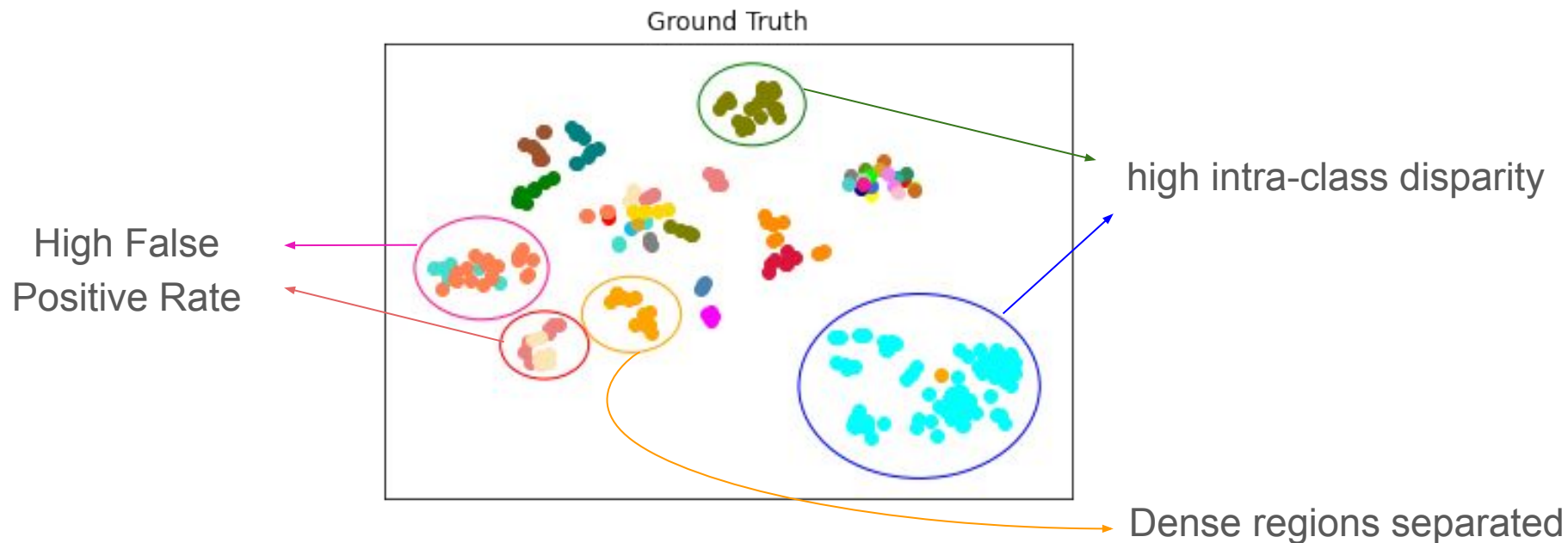


Clustering-based SSL - Leveraging Ensembles

Distance calculation through **Neighborhood distance** and **simple averaging**:



Clustering-based SSL - Leveraging Ensembles



Which is the best density parameter (ϵ) ?

Clustering-based SSL - Leveraging Ensembles

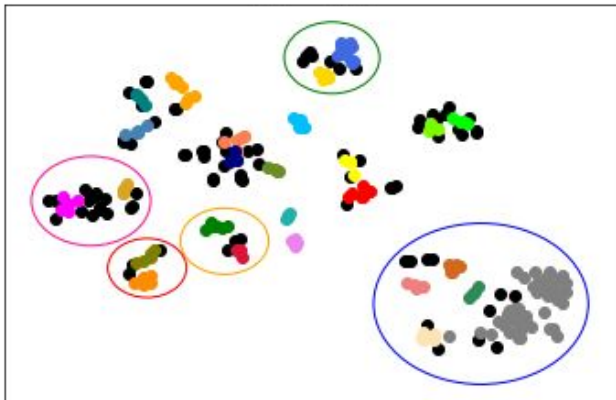
We combine different density-based clustering results

If a sample is an outlier/inlier in the early run, it is kept as outlier/inlier in further ones

The model connects true positive dense regions, remove noise and alleviate ϵ choice

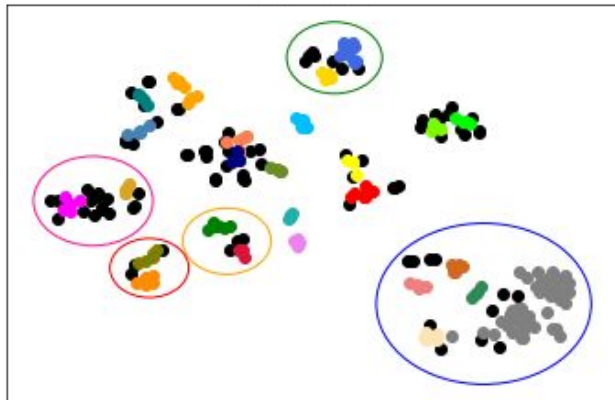
DBSCAN

Eps = 0.50

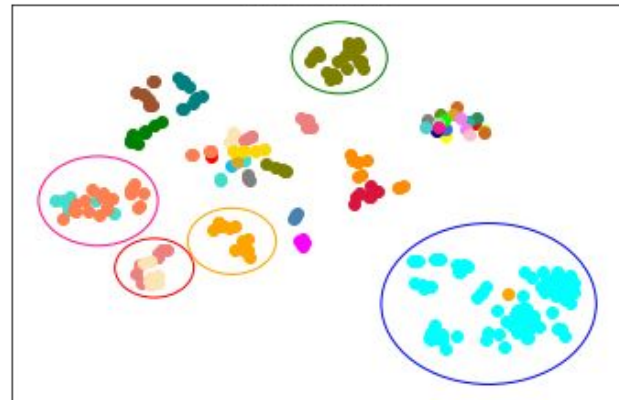


Ours

Eps = 0.50



Ground Truth



Clustering-based SSL - Leveraging Ensembles

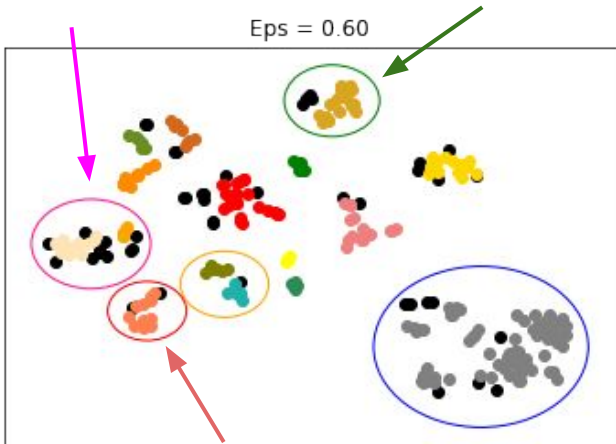
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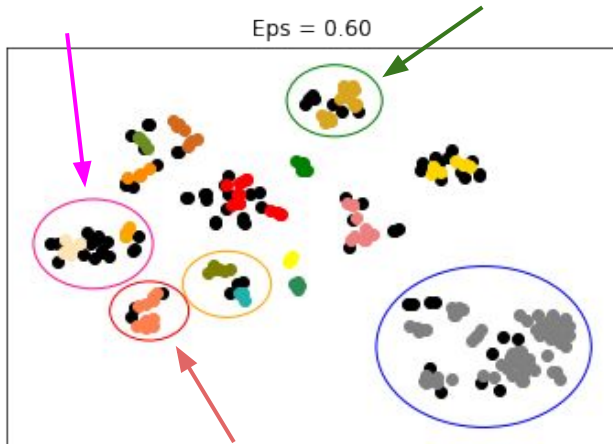
DBSCAN

Eps = 0.60

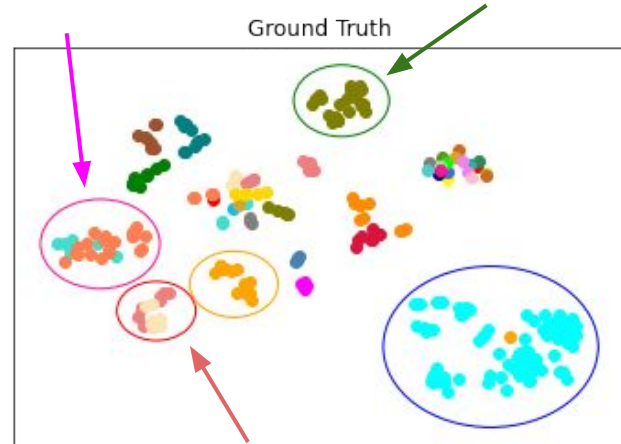


Ours

Eps = 0.60



Ground Truth



Clustering-based SSL - Leveraging Ensembles

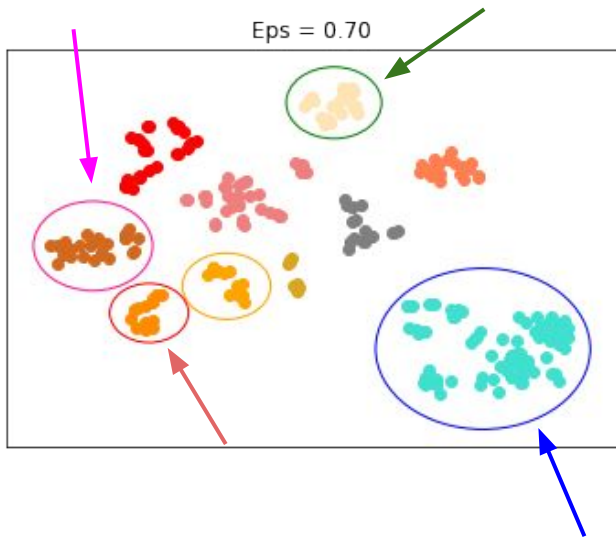
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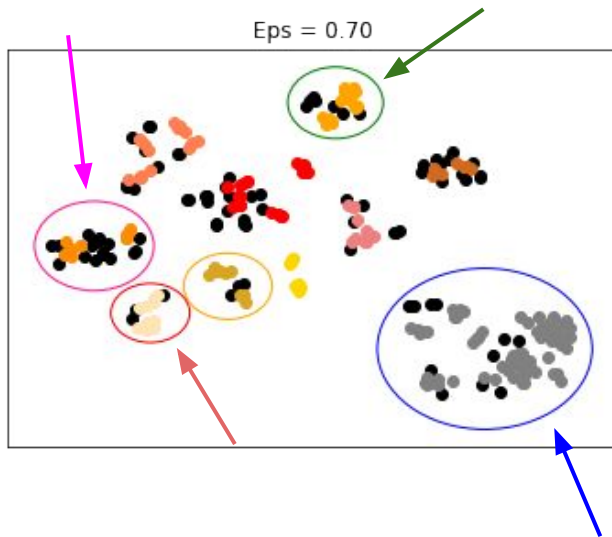
DBSCAN

Eps = 0.70

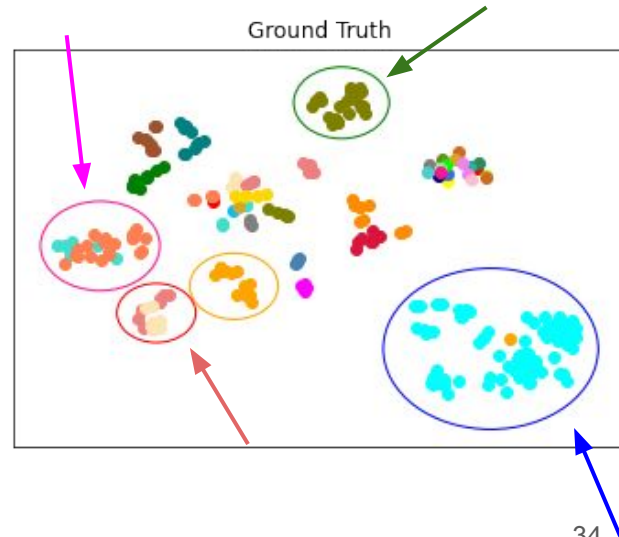


Ours

Eps = 0.70



Ground Truth

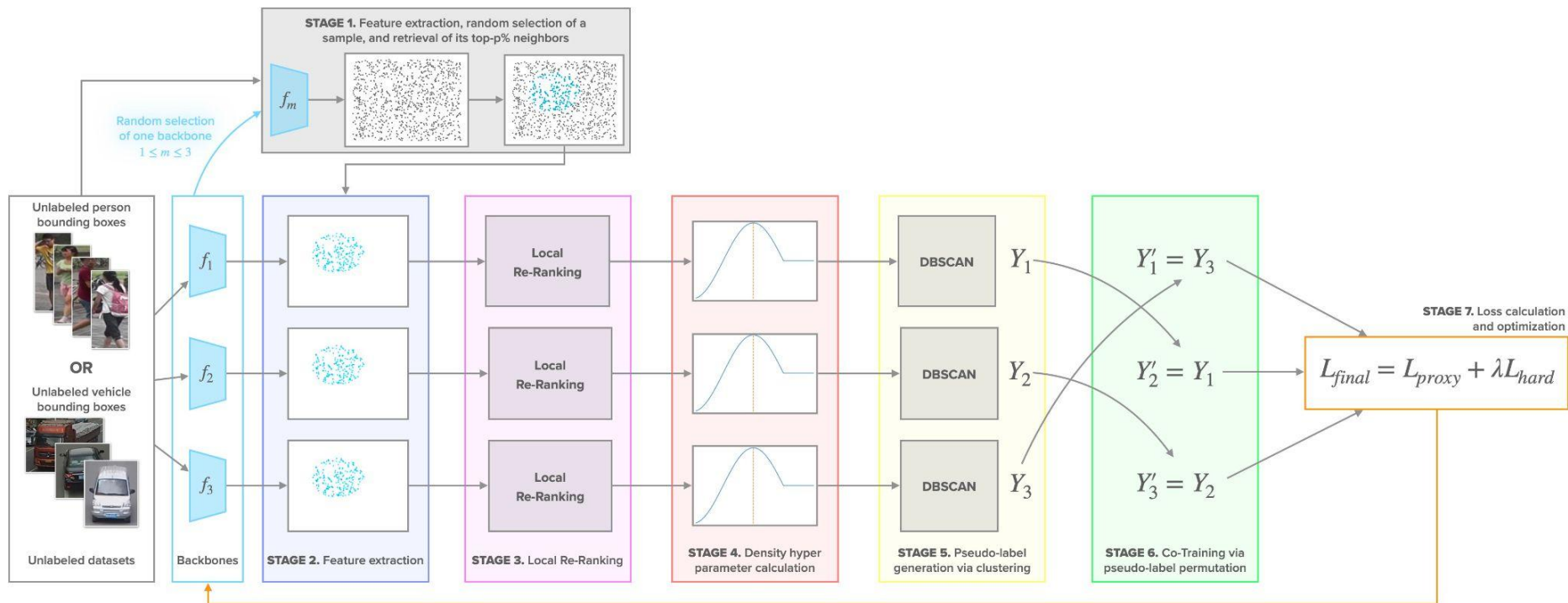


What if other density values are used?

	Market				Duke			
DBSCAN ϵ	mAP	R1	R5	R10	mAP	R1	R5	R10
0.50	81.6	90.7	95.4	96.4	60.9	76.4	82.7	84.7
0.55	82.9	92.3	96.4	97.3	62.8	77.2	83.7	85.8
0.60	82.3	91.6	96.2	97.4	67.5	79.6	87.7	89.9
0.65	82.7	92.3	96.7	97.7	69.1	81.9	88.9	91.5
0.70	81.8	91.8	96.8	97.8	70.3	82.8	90.0	92.1
Ours	83.4	92.9	97.1	97.8	72.7	83.9	91.0	93.0

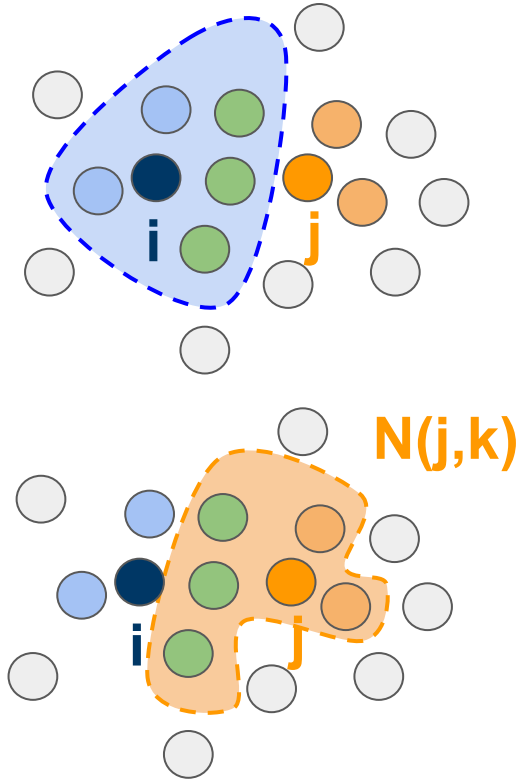
Ours is the BEST!

Clustering-based SSL - Large-scale learning



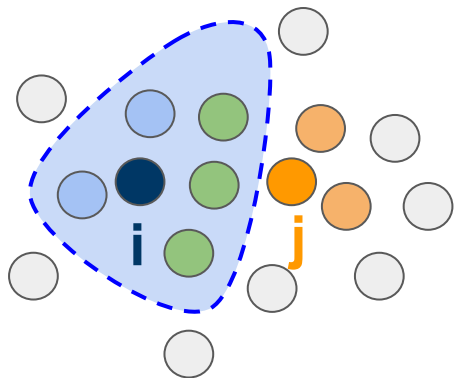
Clustering-based SSL - Large-scale learning

$N(i,k)$



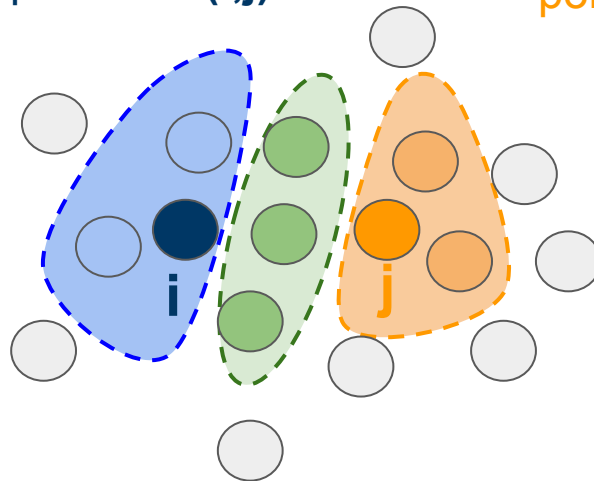
Clustering-based SSL - Large-scale learning

$N(i,k)$



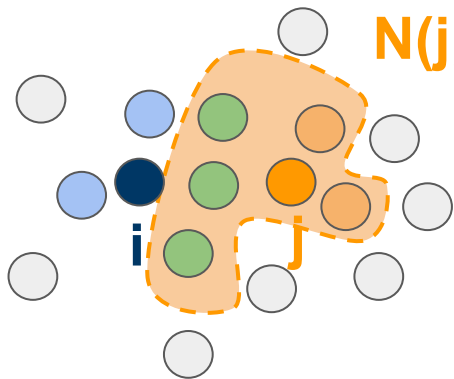
Exclusion set
for point i - $E(i,j)$

Exclusion set for
point j - $E(j,i)$



Inclusion set $I(i,j)$

$N(j,k)$

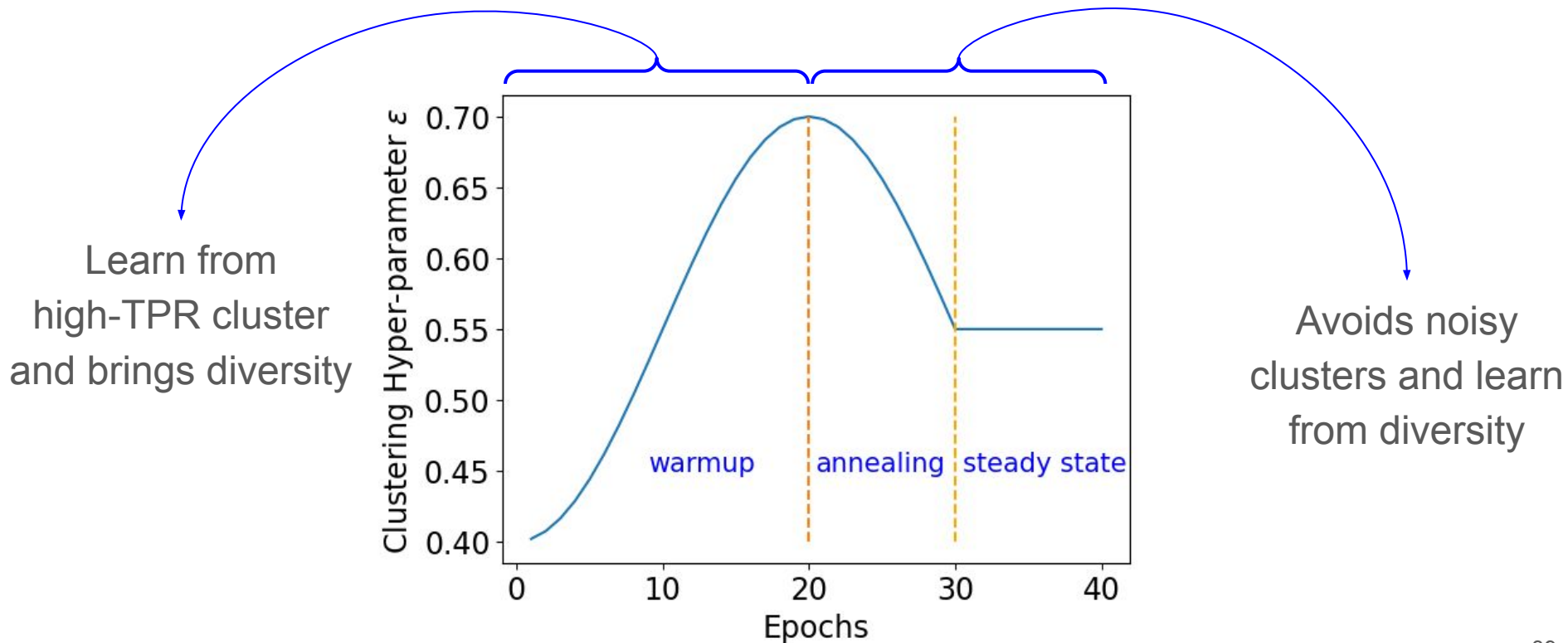


The greater the **NN in common**, the lower is the distance

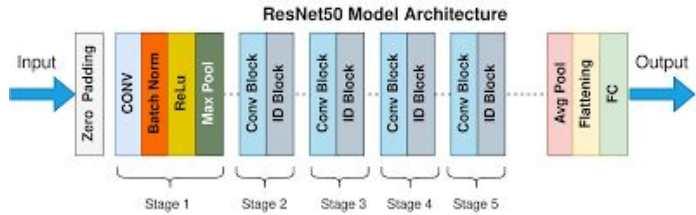
Full Re-Ranking time complexity: $O(n^3)$

Proposed Local Re-Ranking time complexity: $O(n^2 \lg n)$

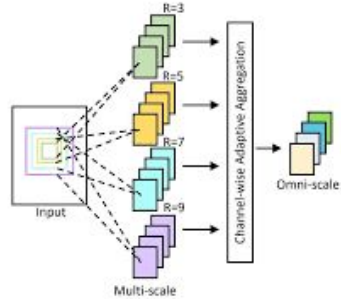
Clustering-based SSL - Large-scale learning



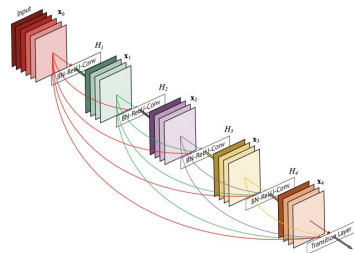
Clustering-based SSL - Large-scale learning



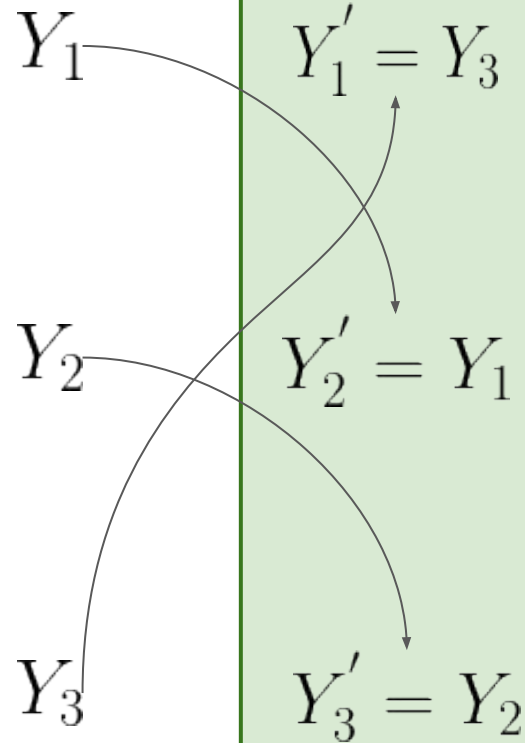
Clustering



Clustering



Clustering



Clustering-based SSL - Large-scale learning

Successful case



Failure case

Multiple strategies (Instance + Clustering) - Dino V2

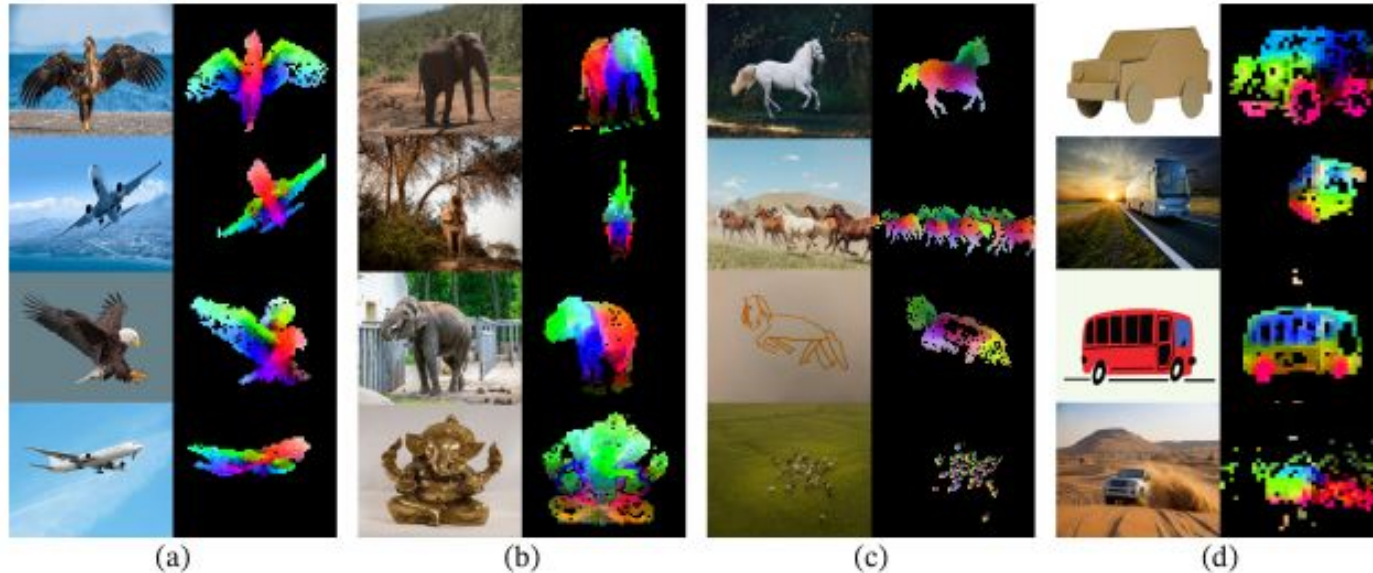
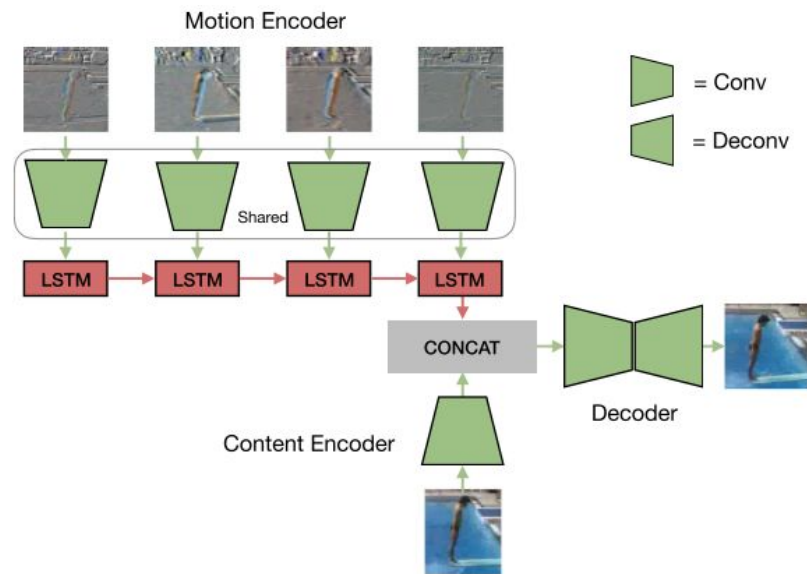
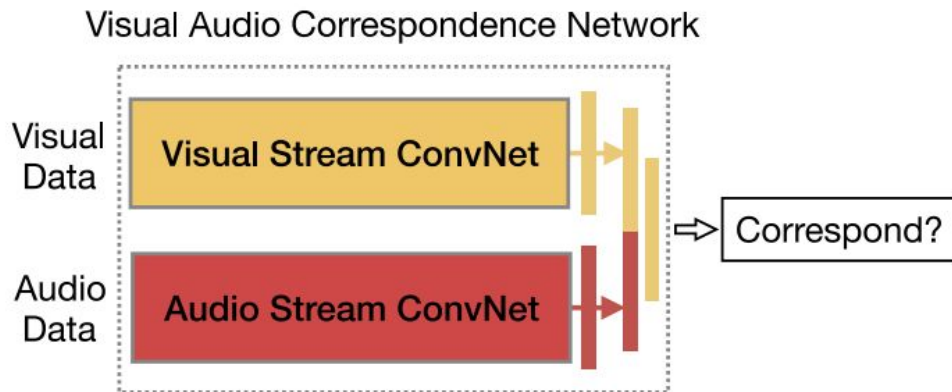


Figure 1: **Visualization of the first PCA components.** We compute a PCA between the patches of the images from the same column (a, b, c and d) and show their first 3 components. Each component is matched to a different color channel. Same parts are matched between related images despite changes of pose, style or even objects. Background is removed by thresholding the first PCA component.

Self-Supervised Learning - Video and multi-modal



SSL in videos

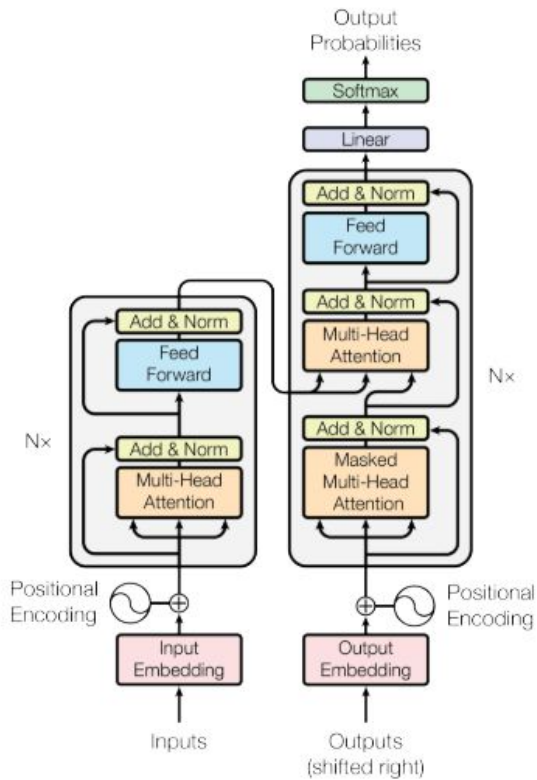


Multi-Modal SSL

Villegas, Ruben, Jimei Yang, Seunghoon Hong, Xunyu Lin, and Honglak Lee. "Decomposing motion and content for natural video sequence prediction." *arXiv preprint arXiv:1706.08033* (2017).

Arandjelovic, Relja, and Andrew Zisserman. "Look, listen and learn." In *Proceedings of the IEEE international conference on computer vision*, pp. 609-617. 43 2017.

Self-Supervised Learning in NLP



Next sentence/word prediction:

“Today is a sunny day! I think I will _____”

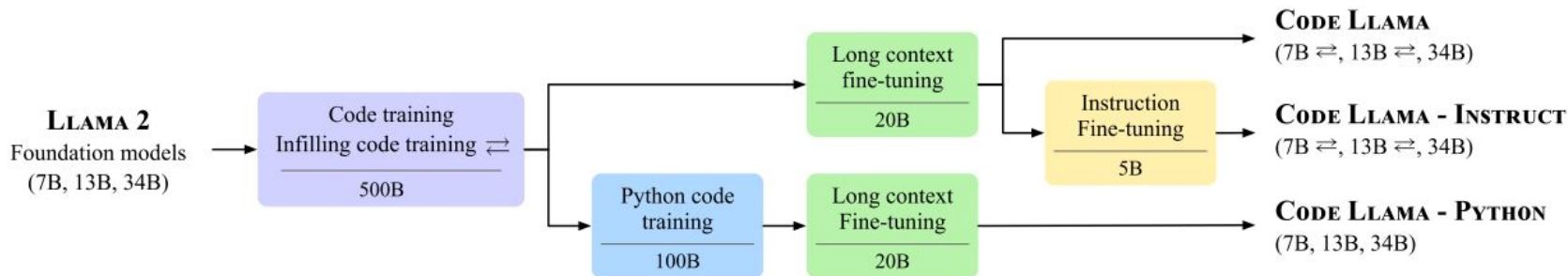
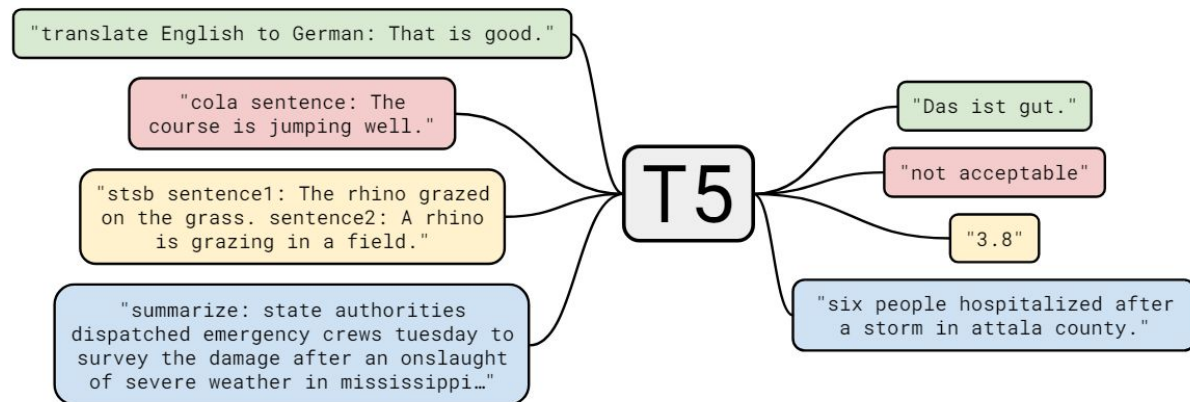
“take a walk”

“take an umbrella”

“take a coat”

95%
2%
3%

Self-Supervised Learning in NLP



Raffel, Colin, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. "Exploring the limits of transfer learning with a unified text-to-text transformer." *Journal of machine learning research* 21, no. 140 (2020): 1-67.

Self-Supervised Learning: Conclusion and Final Remarks

It has been showing promising performance in **varied AI fields** (Biometrics, Forensics, Robotics, Medical imaging, etc.) and giving insights about the feature learning

Recent **breakthrough methods** applied SSL: ChatGPT, GPT-4, Large-Language Models

The trend is to extend its application further and further and **align it with well known strategies**, e.g. supervised learning, reinforcement learning, etc.

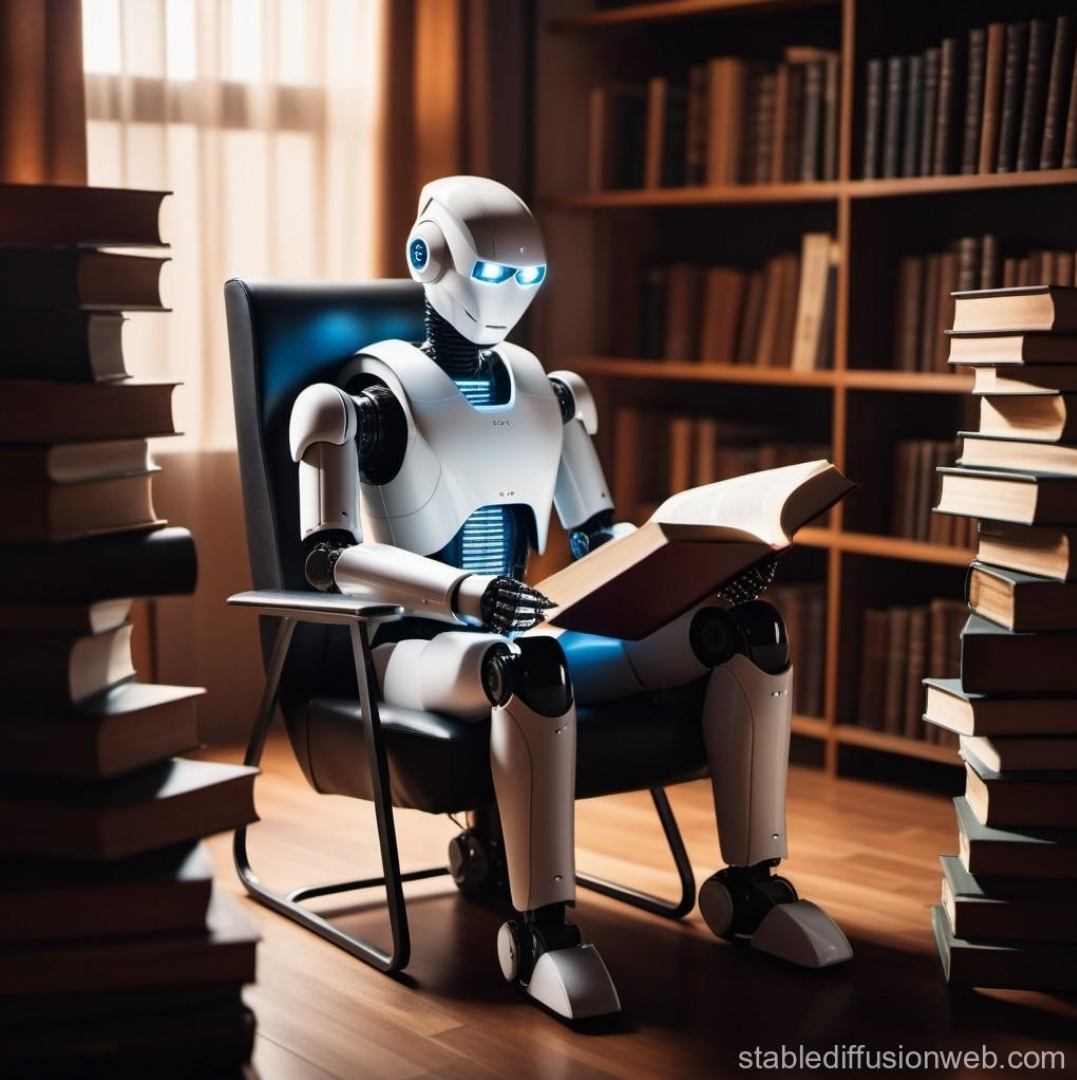
The **cross-modality** has been (and will keep being) extensively explored. The recent **Large Visual Models** employ image and text descriptions for self-supervised learning

In a very close future, maybe will be **impossible to talk about AI without SSL**

Self-Supervised Learning: Conclusion and Final Remarks

Strongly recommended readings:

- Jing, Longlong, and Yingli Tian. "**Self-supervised visual feature learning with deep neural networks: A survey.**" *IEEE transactions on pattern analysis and machine intelligence* 43, no. 11 (2020): 4037-4058.
- Balestrieri, Randall, Mark Ibrahim, Vlad Sobal, Ari Morcos, Shashank Shekhar, Tom Goldstein, Florian Bordes et al. "**A cookbook of self-supervised learning.**" *arXiv preprint arXiv:2304.12210* (2023).
- Sun, Jinghan, Dong Wei, Kai Ma, Liansheng Wang, and Yefeng Zheng. "**Boost supervised pretraining for visual transfer learning: Implications of self-supervised contrastive representation learning.**" In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 2, pp. 2307-2315. 2022.
- Further explanation and examples in Self-Supervised Learning for NLP:
https://cs229.stanford.edu/notes2021spring/notes2021spring/cs229_lecture_selfsupervision_final.pdf



Thank you!